

Roll No.

TEC-659

**B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

ADVANCED EMBEDDED SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain Application Area of Embedded System in detail. (CO1)

OR

- (b) Explain the characteristics of Embedded System. (CO1)

2. (a) Explain the classification of Embedded System. (CO1)

OR

- (b) How can we design a visitor counter using any microcontroller ?
Explain in detail about the design process of it. (CO1)

3. (a) Explain 8051 microcontroller with the help of a block diagram. (CO2)

OR

- (b) What are the addressing modes in the 8051 microcontrollers ? Explain
with suitable examples. (CO2)

P. T. O.

(2)

4. (a) Explain the role of the Program Status Word (PSW) in the 8051 microcontrollers. (CO2)

OR

- (b) Discuss the various Interrupts used in 8051 with the help of an example. (CO3)
5. (a) Write an 8051-assembly language program to toggle a port pin continuously with 1 second delay if crystal frequency is 11.0592 MHz. (CO3)

OR

- (b) Discuss the PIC Microcontroller. (CO3)

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TOE-610

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

OOPs USING C++

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is the difference between Structured Programming and Object Oriented Programming ? (CO1, CO2)

OR

- (b) Write a program in C++ to enter a sentence. Frame a word by joining all the first characters of each word of the sentence. Display the word.

Sample Input : Graphic Era University. (CO1, CO2)

Sample Output : GEU

2. (a) Write a C++ program to implement constructor with default arguments and copy constructor. (CO2, CO3)

OR

- (b) Write a C++ program to overload pre-increment operator (++) by using friend function. (CO2, CO3)

P. T. O.

3. (a) What is the difference between static members and instance members ?
Explain with the help of a C++ program. (CO1, CO2)

OR

- (b) Explain function overloading in C++. Write a C++ program to demonstrate the concept of Compile time polymorphism. (CO1, CO2)
4. (a) With the help of C++ program, explain any *three* uses of scope resolution (::) operator. (CO1, CO3)

OR

- (b) Define a class named TwinData with the following description :
(CO1, CO3)

Data Members :

Private int num1 and num2

Constructor TwinData (int, int) → Constructor will store value in num1 and num 2.

Friend function void twinPrime(TwinData&ob) → This friend function with check whether two numbers are twin prime numbers or not.

Hint : Twin prime numbers are the prime numbers whose difference is 2.

For example : (5, 7), (11, 13), and so on.

5. (a) Write short notes on the following : (CO2, CO3)
- (i) This pointer
 - (ii) Member Access Specifier
 - (iii) Destructor in C++

(3)

OR

(b) Write a C++ program by using class with the following specifications :

(CO2, CO3)

Class name → CharDemo

Private Data Member :

String str → To store the string

Public Member Functions :

void input (string sval) → to assign sval to str

void printInfo() → to check and print the following :

- (i) number of letters
- (ii) number of digits
- (iii) number of uppercase characters
- (iv) number of lowercase characters
- (v) number of special characters

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TEC-602

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
MICROWAVE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) An air-filled rectangular waveguide has a cross section of 6 cm × 4 cm.
- (i) Calculate the cutoff frequency of the dominant mode.
 - (ii) Determine how many modes are passed at three times cutoff frequency of dominant mode.

OR

- (b) In a rectangular waveguide for which $a = 2.5$ cm, $b = 1.0$ cm, $\sigma = 0$,
 $\mu = \mu_0$ and $\epsilon = 4\epsilon_0$. (CO1)

$$H_x = 2 \sin\left(\frac{\pi x}{a}\right) \cos\left(\frac{3\pi y}{b}\right) \sin(\pi \times 10^{11} t - \beta z) \text{ A/m}$$

For TE₁₁ mode, determine :

- (i) The cut-ff frequency
- (ii) The phase constant

P. T. O.

(iii) The propagation constant

(iv) The intrinsic wave impedance

2. (a) Explain attenuation inside the waveguide. Derive the expressions of the surface current components for TE₁₀ mode in a rectangular waveguide having the following field components : (CO1)

$$H_z = A_{10} \cos \frac{\pi x}{a} e^{-j\beta z}$$

$$E_y = \frac{-j\omega\mu a}{\pi} A_{10} \sin \frac{\pi x}{a} e^{-j\beta z}$$

$$H_x = \frac{j\beta a}{\pi} A_{10} \sin \frac{\pi x}{a} e^{-j\beta z}$$

$$E_x = H_y = E_z = 0$$

OR

- (b) If a resonant cavity is filled with a lossless material ($\mu_r = 1$, $\epsilon_r = 2$) and having dimensions $a = 4.5$ cm, $b = 4.2$ cm, and $c = 3.5$ cm is made of copper ($\sigma_c = 5.8 \times 10^7$ S/m). Find the resonant frequency f_r and the quality factor for TE₁₀₁ mode. (CO1)

3. (a) Explain the boundary conditions in a circular waveguide structure to obtain field equations in TE and TM mode. (CO1)

OR

- (b) Find the cut-off frequencies of the first two propagating modes of an air-filled circular waveguide with radius $a = 1.25$ cm. (CO1)

(3)

4. (a) Discuss the different modes of operation in a Gunn diode and mechanism of formation of dipole domain. (CO3)

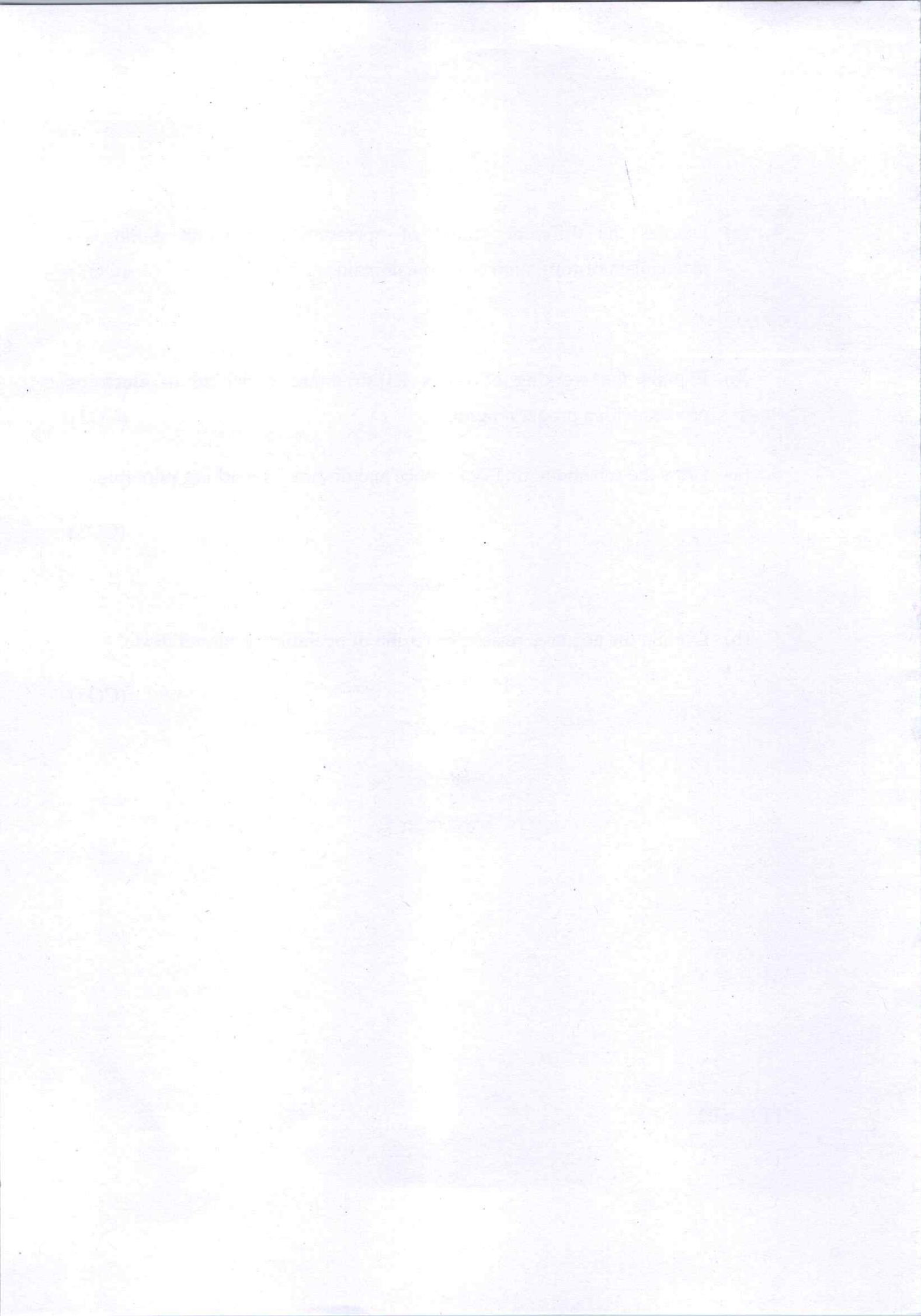
OR

- (b) Explain the working of reflex Klystron and bunching of electrons process with a proper diagram. (CO3)

5. (a) Draw the schematic of TWT device and discuss its working principle. (CO3)

OR

- (b) Explain the negative resistance region of operation in tunnel diode. (CO3)



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TEC-601

B. TECH. (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

WIRELESS COMMUNICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any **one** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Discuss the trends of evolution from 1G to 4 G that took place in Europe and US.

(ii) Define dwell time and the factors on which the dwell time depends.

(CO1)

OR

(b) Verify that the cluster size is $N = i^1 + j^2 + ij$.

2. (a) (i) Find SIR at the boundary of a cell in case of 7, 9, and 12 cells in a cluster. If the minimum required SIR for acceptable voice quality is 18 dB then which one cannot be preferred ? Assume path loss exponent to be 4.

(ii) Explain meaning of the terms 'universal frequency reuse' and 'fractional frequency reuse'.

(CO1)

P. T. O.

OR

- (b) Why sectorization enhances system capacity ? What do you mean by higher order sectorization ? (CO1)
3. (a) If a transmitter produces 50 watt of power uniformly in all direction, express the transmit power in units of (a) dBm, and (b) dBW. If 50 watt is applied to unity gain antenna with a 900 MHz carrier frequency, find the received power in dBm at a free space distance of 100 m from the antenna. What is P_r (10 km) ? Assume unity gain for the receiving antenna. (CO2)

OR

- (b) (i) Derive an expression to represent the received signal using free space path loss model.
- (ii) Assume free space propagation at 900 MHz frequency. Find path loss at a distance of 1 km from the transmitter. (CO2)
4. (a) (i) Discuss different schemes based on the direction of hand-off.
- (ii) Why an umbrella cell approach is needed ? (CO1)

OR

- (b) Assuming 7 cell cluster, find an expression to indicate SIR in co-channel cells in the downlink. (CO2)

(3)

5. (a) (i) A system operating in FDD mode and using a simplex channel of 50 kHz is assigned with 60 MHz frequency spectrum. Find the number of channels per cell if the system uses 7 cells in a cluster.
- (ii) The service area in a cellular system requires 12 repetitions of a cluster having 7 cells. Each cell has 16 channels. Find the total number of channels that can be simultaneously accessed in the system. (CO1, CO2)

OR

- (b) (i) Why a hexagon is preferred to represent a cell ?
- (ii) Define Fraunhofer region. (CO1, CO2)

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TEC-603

B. TECH. (ECE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

VLSI TECHNOLOGY AND DESIGN

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define Scale of Integration, and classification of ICs by structure. (CO1)

OR

- (b) What do you understand by Epitaxy ? Explain Vapor Phase Epitaxy. (CO1)

2. (a) Explain in detail about IC terminology. (CO1)

OR

- (b) Explain in brief about CZ Technology and their application. (CO1)

3. (a) What do you understand by Ion Beam Milling ? Explain its application in VLSI Technology. (CO2)

OR

- (b) Explain the Ion implantation and Ion stopping process and its applications in VLSI. (CO2)

P. T. O.

(2)

4. (a) Explain the various steps of photolithography. (CO2)

OR

- (b) Explain in brief about physical vapor deposition technique for metallization in VLSI. (CO2)

5. (a) Explain Era of VLSI and Moors Law for IC fabrication. (CO3)

OR

- (b) Explain Models of diffusion in solids. (CO3)

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TEC-604

**B. TECH. (ECE) (SIXTH SEMESTER)
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DATA COMMUNICATION NETWORKS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain OSI model with the help of a block diagram. State the addressing, message format used at each layer. (CO1)

OR

- (b) Explain and compare the given network architecture in brief : LAN, WAN and MAN. (CO1)

2. (a) What is packet switching ? Explain virtual circuit network switching with the help of a diagram. (CO1)

OR

- (b) Write short notes on the following : (CO1)

- (i) Sky wave propagation
(ii) Multistage switch

P. T. O.

(2)

3. (a) Explain the services provided by the data link layer. Differentiate between character-oriented and bit-oriented framing. (CO2)

OR

- (b) What are hamming codes ? Discuss its limitations. A data word of at least 11 bits is required. Calculate values of k and n that satisfy this requirement. (CO2)

4. (a) What is High-level Data Link Control (HDLC) protocol ? Explain its frame format and services in detail. (CO2)

OR

- (b) What do you mean by sliding window technique ? Explain Co-back N ARQ protocol. (CO2)

5. (a) Explain the concept of multilevel coding scheme considering 2B1Q coding technique. (CO1, CO2)

OR

- (b) For a (6, 3) systematic LBC, the generator matrix G is given. Construct the code generated by this matrix. Determine the error correcting capabilities of this code Decode the following received word :

(CO1, CO2)

010010

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$